**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 211 · Synoptic retrieval: data structures

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. A three-element array is accessed at index 3. What happens?

Q2. What is the worst-case Big O of a sequential search?

Q3. Push A, B, C. Which pops first?

Q4. In five buckets, h(k)=k MOD 5. Where does 12 start?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

LIFO — Last In, First Out.

FIFO — First In, First Out.

head — Reference to the first node in a linked list.

BST — Binary search tree ordered by key comparisons.

collision — Two hashed keys initially share a bucket.

## New learning

Retrieve operations — A stack pops its newest item; a queue dequeues its oldest item.

Trace changing state — After Push 3, Push 6, Pop, Push 9, the stack is [3,9]. After Enqueue A,B then Dequeue, the queue is [B].

Apply the BST rule — Insert 40 first, 20 left, 60 right, then 30 as the right child of 20.

## Worked example

Trace stack operations Push(2), Push(7), Pop(), Push(5).

1. Predict first — Start with [2,7]. The top of the stack is 7.

2. Check the method — Pop removes 7, leaving [2].

3. Check the method — Push 5 gives [2,5]; the top is now 5.

# We Do · Practise with guidance

Trace queue operations Enqueue(X), Enqueue(Y), Dequeue(), Enqueue(Z).

```
Front → back; remove the oldest item.
```

Predict first

After the two enqueues, the queue is [X,Y].

Check the method

Dequeue removes X, leaving [Y].

Enqueue Z produces [Y,Z]. The next dequeue removes Y.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

A stack contains [4,9] bottom-to-top. Push 2, Pop, Pop. State both removed values and remaining stack.

### Question 2

A queue receives 5,8,3. It dequeues once then enqueues 2. Give the queue and explain why its next dequeue removes 8.

### Question 3

Insert 40,20,60,10,30,25 into a BST. Give the parent of 25, state whether it is a left/right child, and explain what happens to worst-case height when inputs arrive sorted.

# Exit Ticket · Answer before checking

A queue normally removes the most recently enqueued item.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Index 3 is outside indexes 0–2; a C# IndexOutOfRangeException occurs.

Q2. O(n), as it may inspect every element.

Q3. C, because a stack is LIFO.

Q4. Bucket 2.

### You Do pages · Independent questions

Q1. [3] First Pop removes 2 [1]; second Pop removes 9 [1]; remaining bottom-to-top [4] [1].

Q2. [4] Initial front-to-back [5,8,3] [1]; dequeue 5 [1]; enqueue 2 gives [8,3,2] [1]; next removal is 8 because FIFO [1].

Q3. [6] Root 40 [1]; 20 left of 40 [1]; 60 right of 40 [1]; 25 is the left child of 30 [1]. Increasingly sorted input can produce a chain [1], with search up to O(n) [1].

### Exit Ticket · Model answer

False. It removes the oldest enqueued item.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**